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Oryzomys

Oryzomys is a genus of semiaquatic rodents in the tribe Oryzomyini living in southern North America and far northern South America. It includes eight species, two of which—the marsh rice rat (*O. palustris*) of the United States and *O. couesi* of Mexico and Central America—are widespread; the six others have more restricted distributions. The species have had eventful taxonomic histories, and most species were at one time included in the marsh rice rat; additional species may be recognized in the future. The name *Oryzomys* was established in 1857 by Spencer Fullerton Baird for the marsh rice rat and was soon applied to over a hundred species of American rodents. Subsequently, the genus gradually became more narrowly defined until its current contents were established in 2006, when ten new genera were established for species previously placed in *Oryzomys*.

Species of *Oryzomys* are medium-sized rats with long, coarse fur. The upperparts are gray to reddish and the underparts white to buff. The animals have broad feet with reduced or absent ungual tufts of hair around the claws and, in at least some species, with webbing between the toes. The rostrum (front part of the skull) is broad and the braincase is high. Both the marsh rice rat and *O. couesi* have 56 chromosomes, lack a gall bladder, and have a complex penis (as is characteristic of the Sigmodontinae) with some traits that are rare among oryzomyines; these characteristics are unknown in the other species of this genus.

The habitat includes various kinds of wetlands, such as lakes, marshes, and rivers. *Oryzomys* species swim well, are active during the night, and eat both plant and animal food. They build woven nests of vegetation. After a gestation period of 21 to 28 days, about four young are born. Species of *Oryzomys* are infected by numerous parasites and carry at least three hantaviruses, one of which (Bayou virus) also

Oryzomys

Temporal range: Rancholabrean (300,000 years before present) – present



Marsh rice rat (*Oryzomys palustris*)

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Rodentia
Family:	Cricetidae
Subfamily:	Sigmodontinae
Tribe:	Oryzomyini
Genus:	*Oryzomys* Baird, 1857

Type species

Mus palustris Harlan, 1837

Species[1]

- *Oryzomys albiventer*
- *Oryzomys couesi*

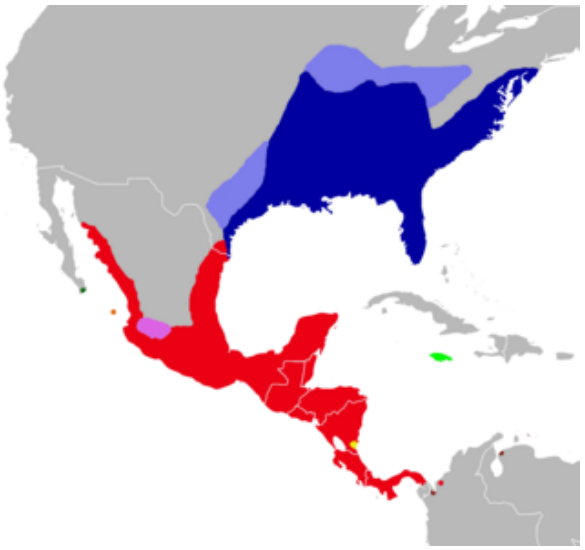
infects humans. Two, maybe three, species have gone extinct over the last two centuries and at least one other is endangered, but the widespread marsh rice rat and *O. couesi* are not threatened.

Taxonomy

Oryzomys is one of about thirty genera within the tribe Oryzomyini, a diverse group of well over a hundred species, many of which were formerly also included in *Oryzomys*.[9] Oryzomyini is one of several tribes within the subfamily Sigmodontinae of the family Cricetidae, which includes hundreds of other species of mainly small rodents, distributed mainly in the Americas and Eurasia.[10]

Within Oryzomyini, a 2006 phylogenetic analysis by Marcelo Weksler which used both morphological and DNA sequence data found some evidence that *Oryzomys* is most closely related to a group including *Holochilus*, *Lundomys*, and *Pseudoryzomys*. Although analyses based on morphological and combined data supported this relationship, sequences of the *Rbp3* gene alone instead placed *Oryzomys* among a group that included *Nectomys*, *Sigmodontomys*, and a few other genera. In all analyses, *Oryzomys* appeared within clade D of Oryzomyini.[11] The relationship between *Oryzomys* and the *Holochilus* group was supported by five synapomorphies (shared derived characters)—absence or reduction of both the hypothenar and interdigital pads; reduction of ungual tufts of hairs surrounding the claws; having the back margin of the zygomatic plate of the skull at the same level as the front of the first upper molar; and the anterocone (front cusp) of the first upper molar divided by an anteromedian fossette. The first three are adaptations to the semiaquatic lifestyle that *Oryzomys* and the members of the *Holochilus* group share, and may thus be examples of convergent evolution.[12]

- *Oryzomys dimidiatus*
- *Oryzomys gorgasi*
- *Oryzomys palustris*
- *Oryzomys peninsulae*
- †*Oryzomys antillarum*
- †*Oryzomys nelsoni*



Distribution of *Oryzomys*: dark blue, marsh rice rat (*O. palustris*); light blue, former distribution of the marsh rice rat; red, *O. couesi*; pink, *O. albiventer*; dark green, *O. peninsulae*; orange, *O. nelsoni*; light green, *O. antillarum*; yellow, *O. dimidiatus* and *O. couesi*; brown, *O. gorgasi*.

Synonyms[1]

- *Hesperomys (Oryzomys)* Baird, 1857[2][Note 1]
- *Oryzomys*: Allen, 1890[3][Note 2]
- *Micronectomys* Hershkovitz, 1948[7][Note 3]

Circumscription

The name *Oryzomys* was introduced in 1857 by Spencer Fullerton Baird for the marsh rice rat (now *Oryzomys palustris*) of the eastern United States,[13] which had been first described twenty years earlier by Richard Harlan.[8] The name combines the Greek *oryza* "rice" and *mys* "mouse" and refers to the feeding habits of the marsh rice rat.[14][15] Baird placed *Oryzomys* as a subgenus of the now-defunct

genus *Hesperomys* and included only the marsh rice rat in it, a classification which was followed by Elliott Coues in 1874 and 1877.[16] In 1890, *Oryzomys* was raised to generic rank, and in subsequent years numerous additional species were ascribed to it, many of which were soon moved to separate genera.[17] In the 1898 *Catalogus Mammalium*, Édouard Louis Trouessart listed 67 species of *Oryzomys*,[18] including some that are now placed in *Calomys*, *Necromys*, *Thomasomys*, and other genera unrelated to *Oryzomys*.[19] Some of the new genera proposed were soon subsumed in *Oryzomys* again,[20] and in *The Families and Genera of Living Rodents* (1941), John Ellerman listed *Microryzomys*, *Oligoryzomys*, *Melanomys*, *Nesoryzomys*, and *Oecomys* as synonyms of *Oryzomys*[21] and included about 127 species in it.[22] In 1948, Philip Hershkovitz suggested that other oryzomyines like *Nectomys* and *Megalomys* could as well be included in *Oryzomys*,[23] and Clayton Ray followed this suggestion in 1962.[24]

Hershkovitz and Ray's classification was never widely followed, and from 1976 on authors started to reinstate some of the other groups lumped in *Oryzomys* as separate genera.[25] The genus was reduced to 43 species (out of 110 in Oryzomyini) in the third edition (2005) of *Mammal Species of the World*,[26] but it was still not a natural, monophyletic group;[27] rather, it mostly united those oryzomyines that lacked the conspicuous specializations of other genera.[28] In 2006, Marcelo Weksler's comprehensive phylogenetic analysis produced further evidence that the genus was polyphyletic, as species of *Oryzomys* were dispersed all over the oryzomyine tree. He proposed that eleven new genera should be created to accommodate those species that were not closely related to the type species of *Oryzomys*, the marsh rice rat;[29] he considered other options that would require fewer new genera, but argued that that would result in less meaningful genus-level groups in Oryzomyini.[30] Later in the same year, Weksler, Percequillo, and Voss created ten new genera—*Aegialomys*, *Cerradomys*, *Eremoryzomys*, *Euryoryzomys*, *Hylaeamys*, *Mindomys*, *Nephelomys*, *Oreoryzomys*, *Sooretamys*, and *Transandinomys*—for species formerly placed in *Oryzomys* and placed six more species related to *"Oryzomys" alfaroi* in *Handleyomys* pending the description of more new genera for them.[31] They left only five species in *Oryzomys*, which was now finally a natural, monophyletic group. Because of subsequent taxonomic work, the number of species has since increased to at least eight.[32]

Some problems remain: ?*Oryzomys pliocaenicus*, a Miocene fossil from Kansas, is of uncertain identity but may belong in *Bensonomys*,[33] and fossils from the Miocene of Oregon and Pliocene of New Mexico have also been ascribed to *Oryzomys*, but probably incorrectly.[33] A possible *Oryzomys* has been recorded from the Irvingtonian (Pleistocene) of Saskatchewan.[34]

Species



Drawing of *Oryzomys molestus*,[35] now a synonym of *Oryzomys albiventer*[36]

The current concept of *Oryzomys* derives from the *palustris-mexicanus* group recognized within a much larger genus *Oryzomys* by Merriam (1901) and the *palustris* group proposed by Goldman (1918).[1] Merriam recognized 21 species within his group, but Goldman consolidated them into eight—the marsh rice rat in the United States, *O. couesi* in far southern Texas, Mexico, and Central America, and six others with small distributions.[37] In 1960, Raymond Hall united *O. couesi* and the marsh rice rat into a single species, *Oryzomys palustris*, and thereafter, other localized forms were also included in *O. palustris*.[38] Hershkovitz described

another species in the group, *O. gorgasi* from Colombia, in 1970[39] and the next year he noted that *O. dimidiatus*, previously classified as a *Nectomys*, was similar to *O. palustris*.[40] After 1979, the marsh rice rat and *O. couesi* were again regarded as separate as a result of further work in Texas, where their ranges meet.[38] While reviewing *O. gorgasi* in 2001, J. Sánchez H. and colleagues redefined and characterized the *O. palustris* group and listed *O. couesi*, *O. dimidiatus*, *O. gorgasi*, and the marsh rice rat as its members;[41] Guy Musser and Michael Carleton in the 2005 third edition of *Mammal Species of the World* additionally listed *O. nelsoni* from María Madre Island in western Mexico.[8]

In 2006, Weksler and colleagues followed the 2001 definition by Sánchez and others for the restricted genus *Oryzomys*, but added *O. antillarum* from Jamaica as a species.[42] Carleton and Joaquin Arroyo-Cabrales reviewed *Oryzomys* from western Mexico in 2009 and in this context provided an extended diagnosis of *Oryzomys*. They recognized eight species: the six previously mentioned plus *O. albiventer* and *O. peninsulae*.[1] Also in 2009, Robert Voss and Weksler identified the subfossil *Oryzomys curasoae* from Curaçao as an island population of *O. gorgasi*.[43] The next year, Delton Hanson and colleagues published a study using DNA sequence data from the cytochrome *b*, interphotoreceptor retinoid-binding protein, and alcohol dehydrogenase 1 genes to assess relationships within *Oryzomys*. They recommended that the marsh rice rat be split into two species and that *O. couesi* be split into four species on the basis of the observed sequence divergence and other data.[44]

Merriam divided his *palustris-mexicanus* group in two "series" according to the color of the underparts (white or fulvous).[45] Goldman divided his *palustris* group in two "sections"—a *couesi* section with *O. couesi* and six related species, and a *palustris* section with *O. palustris* only. He noted that the latter differed from the former in the generally darker, more brownish, longer fur, and larger sphenopalatine vacuities (openings in the mesopterygoid fossa, the gap behind the end of the palate).[46] As Weksler's 2006 analysis included only *O. couesi* and the marsh rice rat among species of *Oryzomys* in the strict sense, he could not test those groups.[47] Carleton and Arroyo-Cabrales concurred with Goldman's division, listing additional characters, and noted that the *palustris* group may be more semiaquatically adapted than the members of the *couesi* group are. In the latter, the fur is usually reddish-brown, as opposed to grayish-brown in the *palustris* group. Members of the *couesi* group have smaller sphenopalatine vacuities and a smaller sphenopalatine foramen, a foramen (opening) in the side of the skull above the molars, and a more highly developed anterolabial cingulum on the third lower molar (a crest at the front of the tooth). The hypothenar pad of the hindfoot, located on the sole far from the fingers, is present in the *couesi* group, but absent in the *palustris* group. Interdigital webbing may be more highly developed in the *palustris* group.[36] Using morphological data, Voss and Weksler found a closer relationship between *O. couesi* and *O. gorgasi* to the exclusion of *O. palustris*, but with low confidence.[48] The DNA sequence data of Hanson and colleagues supported a deep separation between the *palustris* and *couesi* groups, but a Costa Rican sample (assigned to *O. couesi*) was about as distant from the two groups as they were from each other.[49]

The genus currently includes the following species:[1]

Name	Distribution	Characteristics	Taxonomic comments
Oryzomys albiventer	Michoacán, Guanajuato, Jalisco (inland western Mexico)[50]	Large, long tail, robust skull[50]	Described in 1901 as a species; subspecies of *O. couesi* in 1918; reinstated as a species in 2009[51]
Oryzomys antillarum	Jamaica (extinct)[52]	Long nasal bones, short incisive foramina (perforations in the palate), robust zygomatic arches (cheekbones)[53]	Described in 1898 as a species; subspecies of *O. couesi* in 1966; reinstated as a species in 1993/2006[54]
Oryzomys couesi	Southern Texas and coastal Sonora (northwestern Mexico) to northwestern Colombia[55]	Upperparts buff to reddish, underparts white to buff, sphenopalatine vacuities small to absent[56]	Described in 1877 as a species; subspecies of marsh rice rat in 1960; reinstated as a species in 1979; various other species split from and lumped into it from time to time.[57] Genetic data suggest four species may be present: one along the Pacific coast from Sonora to El Salvador; one from Texas to Nicaragua; one in Costa Rica; and one in Panama.[44]
Oryzomys dimidiatus	Southeastern Nicaragua[58]	Gray underparts, brown feet, tail nearly the same color above as below[59]	Described in 1905 as a species of *Nectomys*; reclassified in *Oryzomys* in 1948[60]
Oryzomys gorgasi	Northwestern Colombia, northwestern Venezuela; extinct on Curaçao[61]	Robust rostrum (front part of skull); incisive foramina tapering at the back; sphenopalatine vacuities absent; subsquamosal fenestra (opening at the back of the skull) small[62]	Described in 1970; *Oryzomys curasoae* described in 2001;[58] both synonymized in 2009[43]
Oryzomys nelsoni	María Madre Island, Nayarit, western Mexico (extinct)[63]	Large, long tail, rostrum heavy and bent downward, incisors large and wide[64]	Described in 1898 as a species; subspecies of *O. palustris* in 1971; otherwise regarded as a distinct species[63]
Marsh rice rat (*Oryzomys palustris*)	In the eastern United States from New Jersey and Kansas south to Florida and Texas, and into Tamaulipas, Mexico; previously further north to Iowa and southwestern Pennsylvania[65]	Compared to sympatric *O. couesi* where the two meet: short tail, white underparts, sphenopalatine foramen large[66]	Described in 1827;[63] specific status of Florida Keys form (*argentatus*; first described in 1978) disputed;[67] genetic data suggest populations west of Alabama may be a separate species[68]
Oryzomys peninsulae	Southern tip of Baja California Sur (perhaps extinct)[69]	Moderately large, gray on head and forequarters, broad, squared zygomatic arches, long, broad incisive foramina, upper incisor nearly orthodont[70]	Described in 1899 as a species; subspecies of *O. palustris* in 1971 and of *O. couesi* in 1994; reinstated as a species in 2009[70]

Description

Oryzomys contains medium-sized, semiaquatically specialized oryzomyine rodents. They have long, coarse fur that is grayish to reddish on the upperparts and white to buff on the underparts.[76] The marsh rice rat superficially resembles the introduced species black rat and brown rat, but has larger differences

in color between the upper- and underparts.[75] The vibrissae (whiskers) are short and the ears are small and well-haired. The tail is usually as long as or longer than the head and body and is sparsely haired, but the hairs on the lower side are longer than those above. Females have eight mammae, as in most oryzomyines. The hindfeet are broad and have the first and fifth digits notably shorter than the middle three. The upper surface is hairy, but the underside is naked and covered with small irregularities (squamae). The pads are generally poorly developed, as are the ungual tufts.[77] Interdigital webbing may be present, but its development is variable within the genus.[36]

Measurements of species of *Oryzomys*

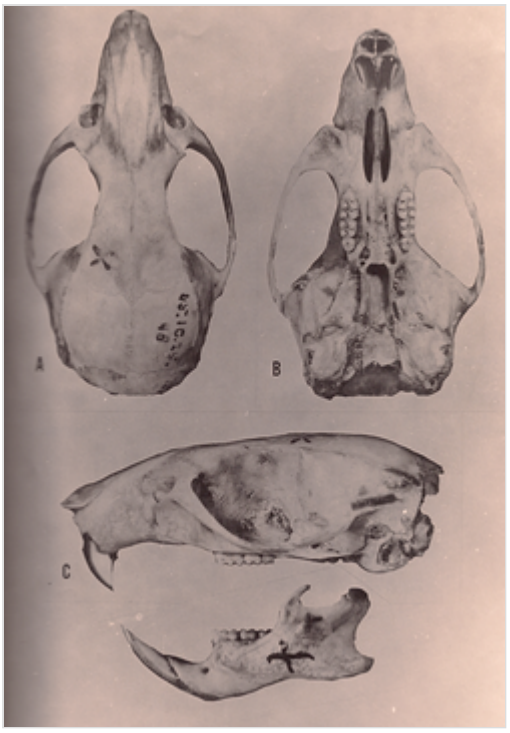
Species	*n*[Note 4]	Total length	Tail	Hindfoot
O. albiventer[71]	12	285.4 (245–314)	155.4 (129–173)	36.1 (33–40)
O. antillarum[72]	3	247 (228–260)	119.7 (108–132)	29.3 (28–30)
O. couesi from Nayarit[71]	62	244.8 (210–288)	125.1 (105–150)	30.5 (27–33)
O. dimidiatus[73]	3	249 (228–278)	129 (110–150)	28.3 (27–31)
O. gorgasi[74]	6–10[Note 5]	259 (220–290)	130 (116–138)	31 (30–32)
O. nelsoni[71]	4	322 (288–344)	181.5 (160–191)	37.3 (35–39)
Marsh rice rat[75]	–	226–305	108–156	28–37
O. peninsulae[71]	14	265.6 (227–305)	136.8 (114–156)	32.0 (29–34)

Measurements are in millimeters and in the form "average (minimum–maximum)".

The karyotype has been recorded in various populations of the marsh rice rat and *O. couesi* and is apparently stable within the genus at 56 chromosomes, with the fundamental number of chromosomal arms ranging from 56 to 60 (2n = 56, FN = 56–60).[44] In both species, the stomach has the characteristic pattern of sigmodontines (unilocular-hemiglandular): it is not split in two chambers by an incisura angularis and the front part (antrum) is covered by a glandular epithelium.[78] Furthermore, the gall bladder is absent, a synapomorphy of Oryzomyini.[79]

Oryzomys species have a large skull with a short rostrum and high braincase. The interorbital region, located between the eyes, is narrowest to the front and is flanked by well-developed beads at its margins. The zygomatic plate is broad and has a well-developed zygomatic notch at its front. The zygomatic arch is robust and contains a small but distinct jugal bone. The interparietal bone, part of the roof of the braincase, is narrow and short;[81] its narrowness is a synapomorphy for *O. couesi* plus the marsh rice rat according to Weksler's analysis.[12] The incisive foramina are long, with their back margin at the front of the first molars or further back. The palate is also long, extending beyond the back margin of the maxillary bone, and is perforated near the third molars by well-developed posterolateral palatal pits. There is no alisphenoid strut, an extension of the alisphenoid bone that in some other oryzomyines separates two foramina in the skull. The auditory bullae are large. The condition of the arteries in the head is highly derived.[81] In the mandible (lower jaw), the coronoid process, a process at the back, is well developed[82] and the capsular process, a raising of the mandibular bone housing the root of the lower incisor, is conspicuous.[46]

As usual in oryzomyines, the molars are pentalophodont (have the mesolophs and mesolophids, accessory crests, well developed) and bunodont, with the cusps higher than the connecting crests.[1] The cusps on the upper molars are arranged in two longitudinal series, not three as in the black and brown rats.[83] The front cusps of the first upper and lower molar (anterocone and anteroconid, respectively) are broad and



Holotype skull of *O. antillarum*, seen from above (A), below (B) and the left (C)[80]

not divided completely by an anteromedian flexus or flexid. Behind the anterocone, the anteroloph (a smaller crest) is complete and separated from the anterocone.[1] On both the second and third lower molars, the anterolophid (a crest on the inner front corner) is present, a putative synapomorphy of the genus.[12] The first molars have additional small roots in addition to the main ones, so that the upper first molar has four and the lower has three or four roots.[1]

As is characteristic of Sigmodontinae, the marsh rice rat and *O. couesi* have a complex penis, with the baculum (penis bone) displaying large protuberances at the sides.[84] The outer surface of the penis is mostly covered by small spines, but there is a broad band of nonspinous tissue.[85] The papilla (nipple-like projection) on the dorsal (upper) side of the penis is covered with small spines, a character these two species share only with *Oligoryzomys* among oryzomyines examined.[86] On the urethral process, located in the crater at the end of the penis,[87] a fleshy process (the subapical lobule) is present; it is absent in all other oryzomyines with studied penes except *Holochilus brasiliensis*.[88] Both traits are recovered as synapomorphies of *O. couesi* plus the marsh rice rat in Weksler's analysis.[12]

Distribution, ecology, and behavior

The range of *Oryzomys* extends from New Jersey in the eastern United States through Mexico and Central America south to northwestern Colombia and east to northwestern Venezuela and Curaçao.[90] Species of *Oryzomys* usually live in wet habitats such as marshes, streams, and mangroves,[1] but both the marsh rice rat and *O. couesi* are also occasionally encountered in drier habitats.[91] They occur or occurred on many continental-shelf islands and one oceanic island, Jamaica; their adeptness at colonizing islands may be caused by their close association with water and frequent occurrence in coastal wetlands.[69] The oldest fossils date to the Rancholabrean of the United States, about 300,000 years ago; although there have been some earlier North American records, those are not in fact referable to *Oryzomys* or even Oryzomyini.[92] Oryzomyines likely evolved in South America east of the Andes; the presence of *Oryzomys* in Central America and other trans-Andean regions is thought to be the result of one of several independent invasions of this region by oryzomyines.[93] Alternatively, *Oryzomys* may have evolved from the Pliocene North American *Jacobsomys*.[94] *O. antillarum* may have reached Jamaica during the last glacial period while sea levels were low.[52]



Oryzomys couesi (above) and *Tylomys panamensis* (below)[89]

Behavior is known mainly from the marsh rice rat and *O. couesi*, with some scattered data from the other species. *Oryzomys* are semiaquatic, spending much time in the water, and otherwise mainly live on the ground;[95] both the marsh rice rat and *O. couesi* are known to be excellent swimmers and will flee into the water when disturbed.[96] Both are also active during the night and build nests of interwoven vegetation, which may be suspended above the water.[97] Breeding may occur throughout the year in both species, but is known to be seasonally variable in the marsh rice rat.[98] In both, gestation takes about 21 to 28 days and litter size is usually one to seven, averaging three to five.[99] Young marsh rice rats and *O. couesi* become reproductively active when about 50 days old.[100]

The marsh rice rat, *O. couesi*, and *O. gorgasi* are known to be omnivores, eating both plant and animal material. They eat both seeds and green plant parts and consume a variety of animals, including insects, crustaceans, and many others.[101] The American barn owl (*Tyto furcata*) is a major predator on the marsh rice rat[102] and remains of *O. antillarum*, *O. couesi*, and *O. gorgasi* have been found in owl pellet deposits.[103] Several other animals are known to prey on *Oryzomys*.[104] A variety of parasites are known from *O. couesi*[105] and the marsh rice rat[106] and two parasitic nematodes have been found in *O. gorgasi*.[107]

Human interactions

Two species of *Oryzomys*, *O. antillarum* and *O. nelsoni*, have gone extinct since the 19th century,[108] and a third, *O. peninsulae*, is unlikely to be still extant.[69] Their extinction may have been caused by habitat destruction and by introduced species such as the small Asian mongoose and the brown and black rat.[109] These same causes may threaten *O. gorgasi*, which the IUCN Red List assesses as "Endangered".[110] *O. albiventer* has been affected by human alteration of its habitat, but likely still survives.[111] In contrast, the widespread species, the marsh rice rat and *O. couesi*, are common and of no conservation concern—indeed, both have been considered a pest—but some populations are threatened.[112] Like these two species, *O. dimidiatus* is assessed as "Least Concern" by the Red List.[113]

The marsh rice rat is the natural reservoir of the Bayou virus, the second most common cause of hantavirus pulmonary syndrome in the United States.[114] Two other hantaviruses, Catacamas virus and Playa de Oro virus, occur in *O. couesi* in Honduras and western Mexico, respectively, but are not known to infect humans.[115]

Notes

1. As a subgenus of *Hesperomys*.
2. First usage as genus.[1] Goldman[4] and Tate[5] attribute the first usage of *Oryzomys* as a genus to Coues, also in 1890.[6]
3. *Nomen nudum* ("naked name").[8]
4. Number of specimens measured.
5. Six for total length, eight for tail length, and ten for hindfoot length.

References

1. Carleton and Arroyo-Cabrales, 2009, p. 116
2. Baird, 1857, p. 482, cf. p. 459
3. Allen, 1890, p. 187
4. Goldman, 1918, p. 11
5. Tate, 1932, p. 4
6. Coues, 1890, p. 4164
7. Hershovitz, 1948, p. 54
8. Musser and Carleton, 2005, p. 1144
9. Weksler, 2006, pp. 1, 10; Weksler et al., 2006, p. 1, table 1
10. Musser and Carleton, 2005
11. Weksler, 2006, figs. 34–39
12. Weksler, 2006, p. 131
13. Baird, 1857, p. 482
14. Schwartz and Schwartz, 2001, p. 192
15. ὄρυζα (https://www.perseus.tufts.edu/hopper/text?doc=Perseus:text:1999.04.0057:entry=o/ruza), μῦς (https://www.perseus.tufts.edu/hopper/text?doc=Perseus:text:1999.04.0057:entry=mu=s). Liddell, Henry George; Scott, Robert; *A Greek–English Lexicon* at the Perseus Project.
16. Coues, 1874, pp. 183–184; 1877, p. 113
17. Tate, 1932, pp. 4ff.
18. Trouessart, 1898, pp. 523–527
19. Tate, 1932, pp. 6–7; Musser and Carleton, 2005, pp. 1105, 1108, 1128, 1130, 1144, 1179, 1180
20. Weksler, 2006, table 1; Musser and Carleton, 2005, p. 1144
21. Ellerman, 1941, p. 340
22. Ellerman, 1940, pp. 345–359
23. Hershkovitz, 1948, p. 54, footnote 1
24. Ray, 1962, pp. 16–26
25. Weksler, 2006, p. 10, table 1; Musser and Carleton, 2005, p. 1144
26. Musser and Carleton, 2005, p. 900
27. Musser and Carleton, 2005, p. 1144; Weksler, 2006, p. 10
28. Weksler, 2006, p. 82
29. Weksler, 2006, p. 75
30. Weksler, 2006, fig. 42, p. 77
31. Weksler et al., 2006, pp. 1–2
32. Carleton and Arroyo-Cabrales, 2009, pp. 115–116; Weksler et al., 2006, table 1
33. Weksler, 2006, p. 87
34. Churcher, 1984, p. 149
35. Eliot, 1904, p. 240
36. Carleton and Arroyo-Cabrales, 2009
37. Goldman, 1918, p. 16
38. Musser and Carleton, 2005, p. 1147
39. Hershkovitz, 1970, p. 700
40. Hershkovitz, 1971, pp. 789, 791
41. Sánchez et al., 2001, pp. 209–210
42. Weksler, 2006, table 1, footnote e

43. Voss and Weksler, 2009, p. 73
44. Hanson et al., 2010, pp. 342–343
45. Merriam, 1901, p. 275
46. Goldman, 1918, p. 20
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48. Voss and Weksler, 2009, fig. 1
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